On 3-fold flips in char $p > 0$.

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Outline of the talk

1. Introduction

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- If X is of general type (i.e. K_X is big so that $h^0(mK_X) = O(m^{\dim X})$), then X has a minimal model which is given by a finite sequence of flips and divisorial contractions $X \dashrightarrow X_1 \dashrightarrow X_2 \ldots \dashrightarrow X_{\min}$. In particular $K_{X_{\min}}$ is nef.

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- If K_X is not pseudo-effective (i.e. $K_X + \epsilon H$ is not big for any H ample and $0 < \epsilon \ll 1$), then after finitely many flips and divisorial contractions we obtain a Mori fiber space: $X_N \to Z$. In particular $-K_{X_N}$ is ample over Z and $\dim X > \dim Z$.

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- Given a normal proj. surface X, $\Delta = \sum \delta_i \Delta_i \geq 0$ s.t.
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  (2) $k = \overline{F}_p$ and $0 \leq \delta_i$ or
  (3) (X, Δ) is LC,
  then there exists a finite sequence of $K_X + \Delta$ negative contractions of irreducible curves
  $X \to \ldots \to X_i \to X_{i+1} \to \ldots X_N$ s.t.
  • X_N is a minimal model ($K_{X_N} + \Delta_N$ is nef) or
  • X_N is a Mori fiber space ($\exists\ f : X_N \to Z$ with $\rho(X_N/Z) = 1$, $\dim Z < \dim X_N$ and $-(K_{X_N} + \Delta_N)$ is f-ample).

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- SLC abundance: If $K_X + \Delta$ is nef, then it is semiample.

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- If there is a proper morphism of algebraic spaces contracting $E(L)$, then L is **EWM** (endowed with map).

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- **Base point free theorem:** If X is normal and $\mathbb{Q}$-factorial, $0 \leq \Delta < 1$, L is a nef and big Cartier divisor such that $L - (K_X + \Delta)$ is nef and big then L is EWM. (If $k = \bar{F}_p$ then L is semiample.)

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- **Cone Theorem:** Assume moreover that $\kappa(K_X + \Delta) \geq 0$ then (1) $\overline{NE}_1(X) = \overline{NE}_1(X)_{(K_X+\Delta)_{\geq 0}} + \sum_{i \in \mathbb{N}} \mathbb{R}[C_i]$,
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- Flips exist (under "mild" restrictions).

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Theorem

(Hacon-Xu) Let (X, Δ) *be a* $\mathbb{Q}$*-factorial* 3*-fold projective dlt pair,* $\Delta \in \{1 - \frac{1}{n} | n \in \mathbb{N}\}$ *and* $p > 5$ *then flips exist.*

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- If the flip exists (assuming Z is affine), then it is given by $X^+ = \mathrm{Proj} \oplus_{m \in \mathbb{N}} H^0(m(K_X + \Delta))$.
- So one must show that $R(K_X + \Delta)$ is finitely generated.

Corollary

If K_X is pseudo-effective and terminal, then it has a minimal model.

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- Similarly if $X \to Z$ is a divisorial contraction and X is projective, then we construct a projective "divisorial contraction" $X' \to Z$ as the relative minimal model of X over Z.
- Base point free theorem, existence of flips in characteristic ≤ 5 or for Δ with arbitrary coefficients, termination of klt flips and abundance are all open for 3-folds.

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2. Extending sections in Char $p > 0$
3. Proof of the existence of flips

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- M is nef if $M \cdot C \geq 0$ for any curve $C \subset X$ and nef and big if moreover $M^{\dim X} > 0$.
- If $H^1(X, \mathcal{O}_X(D)) = 0$, then the restriction $H^0(X, \mathcal{O}_X(D + S)) \to H^0(S, \mathcal{O}_S((D + S)|_S))$ is surjective and we may apply induction on $\dim X$.

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- By duality we identify $\mathcal{H}om(F_*\omega_X, \omega_X) \cong F_*\mathcal{H}om(\omega_X, \omega_X)$, and we let $Tr : F_*\omega_X \to \omega_X$ be the element corresponding to id_{ω_X}.

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- We obtain homomorphisms $\ldots F_*^{e+1}\omega_X \to F_*^e\omega_X \to \ldots F_*\omega_X \to \omega_X$.

Canonical subsystems of adjoint bundles

- If X is smooth and $\mathcal{L}$ is a line bundle, then we also obtain homomorphisms $F^e_*(\omega_X \otimes \mathcal{L}^{p^e}) \cong (F^e_*\omega_X) \otimes \mathcal{L} \to \omega_X \otimes \mathcal{L}$.

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- If $\mathcal{L}$ is ample, then $H^1(X, \omega_X \otimes \mathcal{L}^{p^e}) = 0$ for $e \gg 0$ so that $H^0(X, \omega_X(S) \otimes \mathcal{L}^{p^e}) \to H^0(S, \omega_S \otimes \mathcal{L}|_S^{p^e})$ is surjective.

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- The above result generalizes to log pairs by work of Schwede:

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- If (X, Δ) is a pair such that $(p^e - 1)(K_X + \Delta)$ is Cartier for some $e > 0$ (i.e. p does not divide the index of $K_X + \Delta$). then we get $\Phi^e_\Delta : F^e_* \mathcal{O}_X((1 - p^e)(K_X + \Delta)) \to \mathcal{O}_X$ (by adding $(p^e - 1)\Delta$ and using Tr on the smooth locus).

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- $\sigma(X, \Delta)$ denotes the image of Φ^e_Δ for $e > 0$ sufficiently divisible and $S^0(X, \sigma(X, \Delta) \otimes \mathcal{L})$ the image on global sections after tensoring by $\mathcal{L}$.

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- Analogs of LC and KLT.

Extension Theorem

Theorem (Schwede)

Let (X, Δ) be an F-pure pair such that p does not divide the index of $K_X + \Delta$. Assume that there is a normal non-F-regular center Z so that

$$\Phi^e_\Delta(\mathcal{I}_Z \cdot \mathcal{O}_X((1-p^e)(K_X+\Delta))) \subset \mathcal{I}_Z.$$

If $\mathcal{M}$ is Cartier and $\mathcal{M} - (K_X + \Delta)$ is ample, then

$$S^0(X, \sigma(X, \Delta) \otimes \mathcal{M}) \to S^0(Z, \sigma(Z, \Delta_Z) \otimes \mathcal{M}|_Z)$$

is surjective (where Δ_Z is defined by "adjunction").

Question: Can we arrange $S^0(Z, \sigma(Z, \Delta_Z) \otimes \mathcal{M}|_Z) \neq 0$?

Global F-regularity

- If $\mathcal{M}$ is sufficiently ample, then $S^0(X, \sigma(X, \Delta) \otimes \mathcal{M}) \cong H^0(X, \sigma(X, \Delta) \otimes \mathcal{M})$ is globally generated (in general the inclusion can be strict!).

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- If $\mathcal{M}$ is big and semiample, let $f : X \to T = \mathrm{Proj}(\oplus_{m \geq 0} \mathcal{M}^m)$ Then (replacing $\mathcal{M}$ by a multiple)

$$H^0(X, \mathcal{O}_X((p^e - 1)(K_X + \Delta)) \otimes \mathcal{M}^{p^e}) \to H^0(X, \mathcal{M})$$

is surjective provided that (X, Δ) is F-regular over T i.e. $F^e_* f_* \mathcal{O}_X((p^e - 1)(K_X + \Delta)) \to \mathcal{O}_T$ is surjective.

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- By a result of Schwede-Smith X/T is of relative log Fano type.

Global F-regularity II

- If $\dim X = 2$, $p > 5$, (X, Δ) is klt, $\Delta \in \{1 - \frac{1}{n} | n \in \mathbb{N}\}$, $f : X \to T$ is birational and $-(K_X + \Delta)$ is f-ample then (X, Δ) is globally F-regular over T.

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- The proof is by "classification"; uses Shokurov's theory of complements.
- We will now give a sketch of the proof of the existence of 3-fold flips which closely follows ideas of Shokurov.

Outline of the talk

Pl-flips and the restricted algebra

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- Let $R_S(K_X + \Delta) = \mathrm{Im}(R(K_X + \Delta) \to R(K_{S^n} + B_{S^n}))$ where $S^n \to S$ is the normalization and $K_{S^n} + B_{S^n} = (K_X + \Delta)|_{S^n}$.

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- Then $R(K_X + \Delta)$ is fin. gen. iff $R(K_{S^n} + B_{S^n})$ is fin. gen.
- We have $K_X + \Delta = tS$; assume $t = 1$, then the kernel of $H^0(m(K_X + \Delta)) \to H^0(m(K_{S^n} + B_{S^n}))$ is in $H^0((m-1)(K_X + \Delta))$.

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- If $R_S(K_X + \Delta) = R(K_{S^n} + B_{S^n})$ we would be done! So we must identify which elements of $R(K_{S^n} + B_{S^n})$ lift.

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- Pick H sufficiently ample on X ($H|_{S^n}$ very ample).

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- As $K_{S^n} + B_{S^n}$ is F-regular, we have a surjection $F^e_*\mathcal{O}_{S^n}((1-p^e)(K_{S^n} + B_{S^n} + \delta E + p^e H|_{S^n})) \to H^0(\mathcal{O}_{S^n}(H))$.

The restricted algebra

- For any $f : Y \to X$, let $N_{i,Y} = \mathrm{Mob}(i(K_X + \Delta))$ and $M_{i,S'} = N_{i,Y}|_{S'}$, $D_{i,S'} = \frac{1}{i} M_{i,S'}$ and $D_{S'} = \lim D_{i,S'}$.

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- For Y sufficiently high, $N_{i,Y}$ and $M_{i,S'}$ are free and $\lceil A_Y \rceil$ (resp. $\lceil A_{S'} \rceil$) saturated i.e. $|N_{i,Y} + \lceil A_Y \rceil| = |N_{i,Y}| + \lceil A_Y \rceil$. (This is because $\lceil A_Y \rceil \geq 0$ is exceptional and so $|f^*(i(K_X + \Delta)) + \lceil A_Y \rceil| = |f^*(i(K_X + \Delta))| + \lceil A_Y \rceil$.)

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- $R_S(K_X + \Delta)$ is finitely generated iff
  • there exists $\bar{S} \to S$ such that $M_{i,S'}$ descends to $\bar{S}$ for all $i \gg 0$ (given $h : S' \to \bar{S}$, then $M_{i,S'} = h^* M_{i,\bar{S}}$ and
  • an index $\bar{i}$ s.t. $D_{i,\bar{S}} = D_{\bar{i},\bar{S}}$ whenever $\bar{i}|i$.

The restricted algebra

- For any $f : Y \to X$, let $N_{i,Y} = \mathrm{Mob}(i(K_X + \Delta))$ and $M_{i,S'} = N_{i,Y}|_{S'}$, $D_{i,S'} = \frac{1}{i} M_{i,S'}$ and $D_{S'} = \lim D_{i,S'}$.
- For Y sufficiently high, $N_{i,Y}$ and $M_{i,S'}$ are free and $\lceil A_Y \rceil$ (resp. $\lceil A_{S'} \rceil$) saturated i.e. $|N_{i,Y} + \lceil A_Y \rceil| = |N_{i,Y}| + \lceil A_Y \rceil$. (This is because $\lceil A_Y \rceil \geq 0$ is exceptional and so $|f^*(i(K_X + \Delta)) + \lceil A_Y \rceil| = |f^*(i(K_X + \Delta))| + \lceil A_Y \rceil$.)
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  • an index $\bar{i}$ s.t. $D_{i,\bar{S}} = D_{\bar{i},\bar{S}}$ whenever $\bar{i}|i$.
- $\mu : \bar{S} \to S$ is just the terminalization of (S, B_S) i.e. the "smallest" resolution such that $\mathrm{mult}_x(B_{\bar{S}}) < 1$ for any $x \in \bar{S}$ and $K_{\bar{S}} + B_{\bar{S}} = \mu^*(K_S + B_S)$ (nb. $B_{\bar{S}} = -A_{\bar{S}} \geq 0$).

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- Since C is an affine curve, the RHS is generated, but the LHS vanishes along $\lceil A_Y \rceil$ and so $C \cap \lceil A_{S'} \rceil = 0$, i.e. $M \cap \lceil A_{S'} \rceil = \emptyset$ as required.

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- Let $a : \bar{S} \to S^+$ be the induced morphism. We claim that $D_{\bar{S}} = a^* D_{S^+}$ where $D_{S^+} \in \mathrm{Div}_{\mathbb{Q}}(S^+)$.

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- Proceeding as above one checks that $(\lceil \frac{j}{i} M_{i,S'} + A_{S'} \rceil - M_{j,S'}) \cdot g^* C = 0$ and hence $\lceil \frac{j}{i} M_{i,S'} + A_{S'} \rceil - M_{j,S'}$ is exceptional over S^+.

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- Finally, the usual saturation arguments (with Kawamata Viehweg vanishing replaced by the S^0 extension results) show that $\frac{1}{i} M_{i,\bar{S}} = D_{\bar{S}}$ for all $i > 0$ sufficiently divisible.